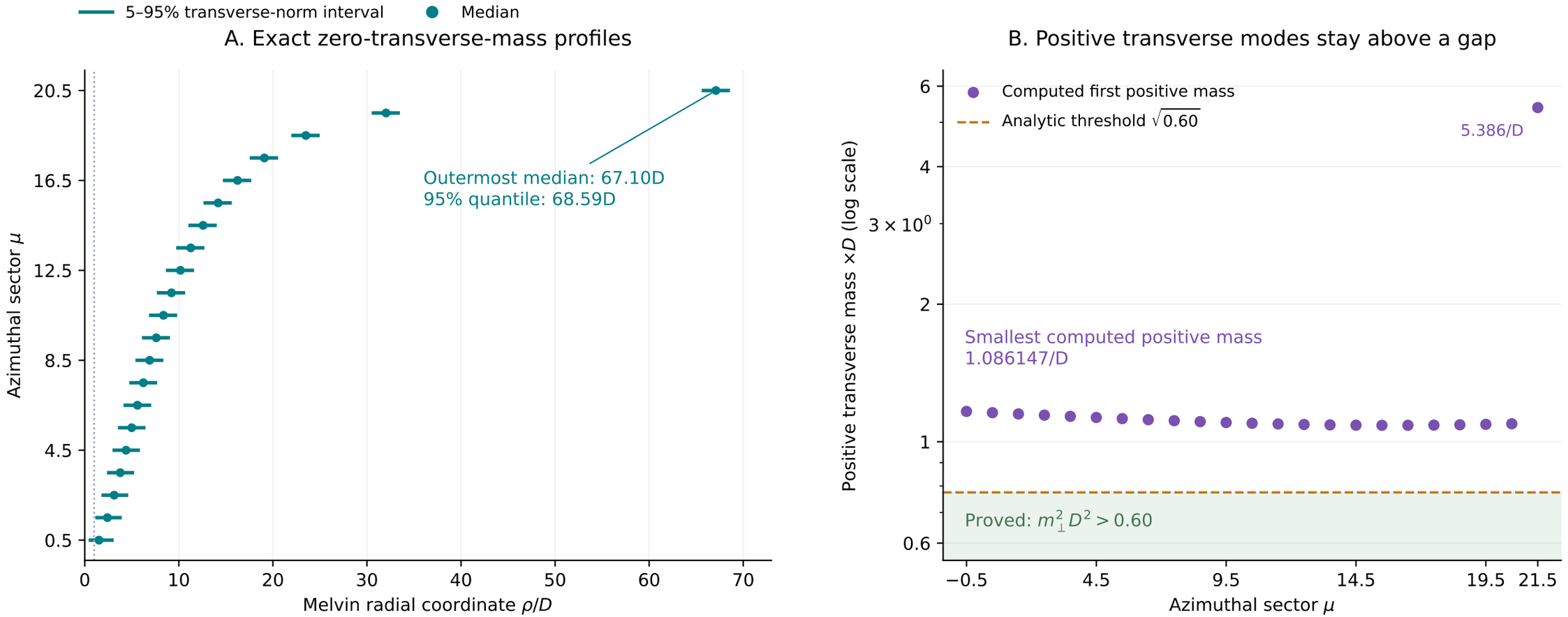
Melvin exterior: 21 gapless channels per Weyl component

Massless charged Dirac field in the asymptotic pure Melvin model · Q = 0.2, D = q = 1 · Mouth matching remains incomplete



Norm intervals show radial spread, not uncertainty or hard walls. These are coordinate radii, not proper transverse distances.

For $|E|D < \sqrt{0.60}$, only the exact zero channels can propagate axially in the pure Melvin model.

Numerical positive masses and the separate analytic interval bound have different roles. Zero masses are excluded from the logarithmic plot.

The actual curved exterior still needs finite-interface matching. Existing horizons remain; no throat, quantum supporting source or stability is established.

13 Sep 2027 → 12 Sep 2027 remains unproved; these channels do not establish a physical medium or time-travel route.